

PBI Title

Retriever Part Exploration and Implementation for Graph RAG

PBI Description

- Explore and study different retriever approaches for RAG and Graph RAG.
- Implement a graph-based retriever using the existing Knowledge Graph.
- Evaluate the retriever using sample queries and improve retrieval relevance.
- Compare vector search and Knowledge Graph-based retrieval to identify the suitable approach.

Acceptance Criteria

- Relevant retriever research is completed and key findings are documented.
- A graph-based retriever is implemented and integrated with the existing Graph RAG pipeline.
- The retriever is tested with sample queries and provides relevant context.
- Vector search and Knowledge Graph retrieval are compared, and the findings are documented.

Tasks

1. Explore Retriever Research Papers

- Study research papers related to RAG and Graph RAG retrievers.
- Understand different retrieval approaches such as vector search, graph traversal, hybrid retrieval, and multi-hop retrieval.
- Identify the most suitable approach for the current Graph RAG use case.

2. Retriever Implementation Based on Knowledge Graph

- Design and implement a graph-based retriever using the existing knowledge graph.
- Retrieve relevant nodes and connected nodes based on the user query.
- Implement multi-hop graph traversal to retrieve related context.
- Integrate the retrieved graph context into the RAG pipeline.

3. Retriever Evaluation and Optimization *(Suggested)*

- Create sample queries to evaluate the retriever.
- Check whether the retrieved nodes and relationships are relevant to the query.

- Identify retrieval gaps and improve the retrieval logic.
- Optimize the retriever for better relevance and efficient retrieval.

4. Comparison of Vector Search and Knowledge Graph Retrieval

- Compare vector-based retrieval with knowledge graph-based retrieval.
- Evaluate both approaches using the same set of sample queries.
- Compare the relevance, context quality, multi-hop retrieval capability, and retrieval performance.
- Document the observations and identify where each approach works best.